generator (fresh process, timed)
a_j = splitmix64(seed) mod r, $R_j = [a_j]G$
orbits $\pm\varphi^s(R_j)$, labels $\pm\lambda^s$; logs $L_j = a_j$

→

baseline producer (unchanged, timed)
retained header + `KIC_LOG_TABLE_IN`
no rank stage, no LA (LAnone)
verifies every $[L_j]G = R_j$, builds the index

→

one-target online
(the baseline's S3 root index scan)
⇒ scalar, oracle replay

E2E(cL) per point = gen_ms (generator's clock) + cold_in_process_ms (producer: base load, normal basis, index, log verification, self-tests, online). Nothing cached. Before timing, `check_knownlog.py` (pinned pure-Python oracle, no shared code) verifies every log, point, label and hash of both bases, and detects three planted corruptions.

cold segments C1..C40: one fresh pinned process per arm on new point s$(i-1)$ (role knownlog-cold-v1, disjoint from every n=83 panel)

cP	**cL600**	**cL220**	cLaa	**rT**
IC baseline cold	known-log K600	known-log K_{opt}	A/A of cL220	tuned rho rT770

kP
IC baseline rank stage

rank blocks K1..K4 (new seeds 901..904), interleaved after every 10 cold segments; factor-log table must equal B's. Q0 (O1): cP, cL600, cL220, cLaa reproduce frozen identities; **cLX** (planted) feeds the producer a table with one log off by one: it must refuse it.

Primary (frozen; a reference measurement, not a test): $G = \overline{\mathrm{E2E}}(\mathrm{cL220}) \,/\, \overline{\mathrm{cold}}(\mathrm{rT})$ with 95% paired point bootstrap, and S of cL220. Reading: above / below / not separated from rho. **Secondary, labelled "reference, not an IC improvement":** $\left(\overline{\mathrm{rank}}(\mathrm{kP}) + \overline{\mathrm{cold}}(\mathrm{cP})\right) / \overline{\mathrm{E2E}}(\mathrm{cL220})$.

Without relations this is generic BSGS on the Frobenius quotient, bounded by the DKM non-adaptive line: it can never beat rho. It never replaces the IC baseline row.